Chapter 4: Continuity

Section: Monotonic Functions

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Motivation

Why Monotonic Functions?

- Functions that never decrease (or never increase) on an interval

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- One-sided limits always exist — no wild oscillation possible
- Discontinuities are at most countable — and can be dense!

Definition

Definition 4.28: Monotonic Functions

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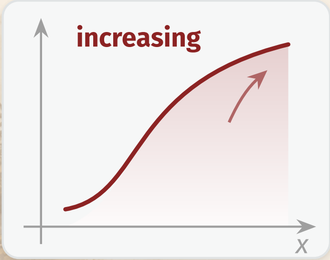
Let f be real on (a, b).

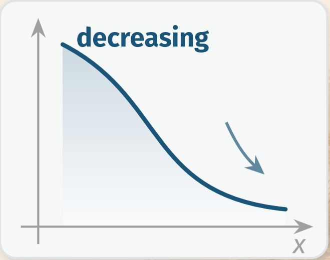
- f is monotonically increasing if $a < x < y < b \Rightarrow f(x) \leq f(y)$.
- f is monotonically decreasing if $a < x < y < b \Rightarrow f(x) \geq f(y)$.

The *monotonic* functions are the increasing ones together with the decreasing ones.

Two Pictures



$x < y \Rightarrow f(x) \leq f(y)$



$x < y \Rightarrow f(x) \geq f(y)$

One-Sided Limits

Theorem 4.29: One-Sided Limits Exist

Theorem 4.29

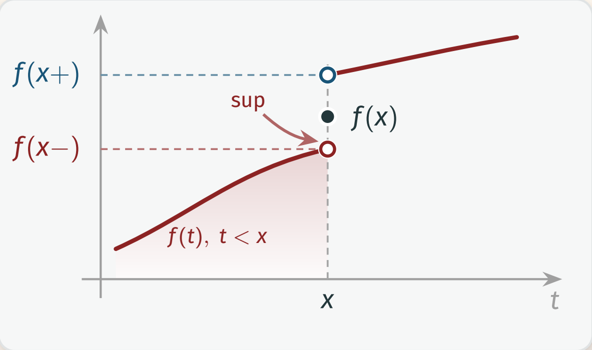
Let f be monotonically increasing on (a, b). Then $f(x+)$ and $f(x-)$ exist at every $x \in (a, b)$:

$$\sup_{a<t<x} f(t) = f(x-) \leq f(x) \leq f(x+) = \inf_{x<t<b} f(t). \tag{25}$$

Furthermore, if $a < x < y < b$, then

$$f(x+) \leq f(y-). \tag{26}$$

Picture: The Left-Hand Limit as a Supremum



Proof of 4.29 — Step 1: Name the Supremum

Step 1: Set up

Values $f(t)$ for $a < t < x$ are bounded above by $f(x)$. Let

$$A = \sup_{a<t<x} f(t), \qquad A \le f(x).$$

Goal: show $A = f(x-)$.

Proof of 4.29 — Step 2: Use the Supremum

Step 2: Approximate from below

Given $\varepsilon > 0$, by definition of A there is $\delta > 0$ with $a < x - \delta < x$ and

$$A - \varepsilon < f(x - \delta) \leq A. \tag{27}$$

Proof of 4.29 — Step 3: Squeeze with Monotonicity

Step 3: Trap all nearby values

For $x - \delta < t < x$, monotonicity gives

$$f(x-\delta) \le f(t) \le A. \tag{28}$$

Combining (27) and (28): $|f(t) - A| < \varepsilon$ for $x - \delta < t < x$.

$$\boxed{f(x-) = A = \sup_{a<t<x} f(t)}$$

Proof of 4.29 — Step 4: Comparing Across Points

Step 4: Why $f(x+) \leq f(y-)$

For $a < x < y < b$:

$$f(x+) = \inf_{x<t<b} f(t) = \inf_{x<t<y} f(t), \tag{29}$$

$$f(y-) = \sup_{a<t<y} f(t) = \sup_{x<t<y} f(t). \tag{30}$$

Every value on (x, y) lies between these $\Rightarrow f(x+) \leq f(y-)$.

Corollary: No Second-Kind Discontinuities

Corollary

Monotonic functions have no discontinuities of the second kind.

Recall

A discontinuity of the second kind is one where a one-sided limit *fails to exist*. Theorem 4.29 guarantees both always exist.

Countably Many Discontinuities

Theorem 4.30: Discontinuities Are Countable

Theorem 4.30

Let f be monotonic on (a, b). Then the set of points of (a, b) at which f is discontinuous is at most countable.

Strategy: attach a distinct rational number to each jump.

Proof of 4.30 — Step 1: Insert a Rational in Each Jump

Step 1: Define $r(x)$

Take f increasing; let E be its set of discontinuities. At each $x \in E$ we have $f(x-) < f(x+)$, so choose a rational $r(x)$ with

$$f(x-) < r(x) < f(x+).$$

Proof of 4.30 — Step 2: The Map Is Injective

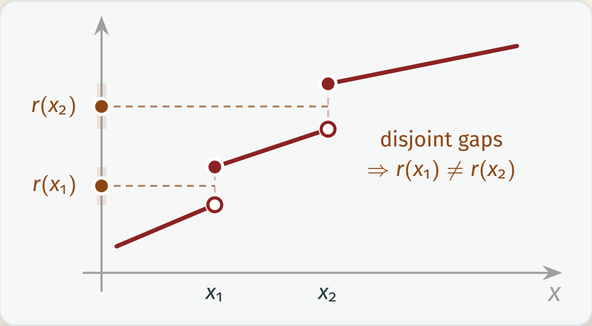
Step 2: Distinct points get distinct rationals

If $x_1 < x_2$, then by (26): $f(x_1+) \leq f(x_2-)$. Hence the jump intervals are disjoint and

$$r(x_1) < f(x_1+) \leq f(x_2-) < r(x_2) \;\Rightarrow\; r(x_1) \neq r(x_2).$$

$$\boxed{E \hookrightarrow \mathbb{Q}, \quad \text{so } E \text{ is at most countable.}}$$

A Picture of Disjoint Jumps



Discontinuities Can Be Dense

Remark 4.31: Jumps Need Not Be Isolated

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Given *any* countable set $E \subset (a, b)$ — even a dense one — there is a monotonic f discontinuous exactly at the points of E.

"At most countable" is sharp: countably many jumps can fill the line densely.

Remark 4.31 — The Construction

Build f from a convergent series

Enumerate $E = \{x_n\}$. Pick $c_n > 0$ with $\sum c_n$ convergent. Define

$$f(x) = \sum_{x_n < x} c_n \qquad (a < x < b). \tag{31}$$

Sum over all n with $x_n < x$ (empty sum $= 0$).

Remark 4.31 — Properties of f

The function f from (31) satisfies

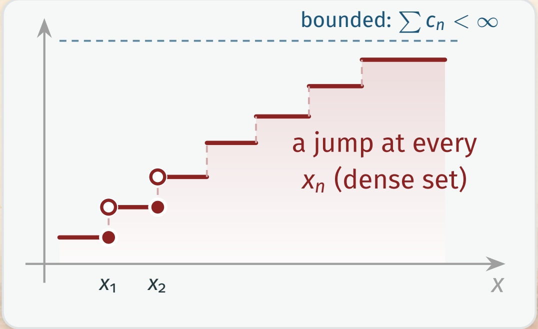
(a) f is monotonically increasing on (a, b);

(b) f is discontinuous at each x_n, with jump $f(x_n+) - f(x_n-) = c_n$;

(c) f is continuous at every other point.

Also $f(x-) = f(x)$ everywhere: f is continuous from the left.

Picture: A Dense Set of Tiny Jumps



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- Def 4.28: monotonic = never decreasing or never increasing
- Thm 4.29: both one-sided limits exist (sup / inf); no second-kind discontinuities
- Thm 4.30: discontinuities are at most countable — but (4.31) can be dense

Coming up next: Infinite Limits and Limits at Infinity.